

$$\mathbb{Z}/d\mathbb{Z} \times \mathbb{F}_{p^d} \longrightarrow \mathbb{F}_{p^d}$$

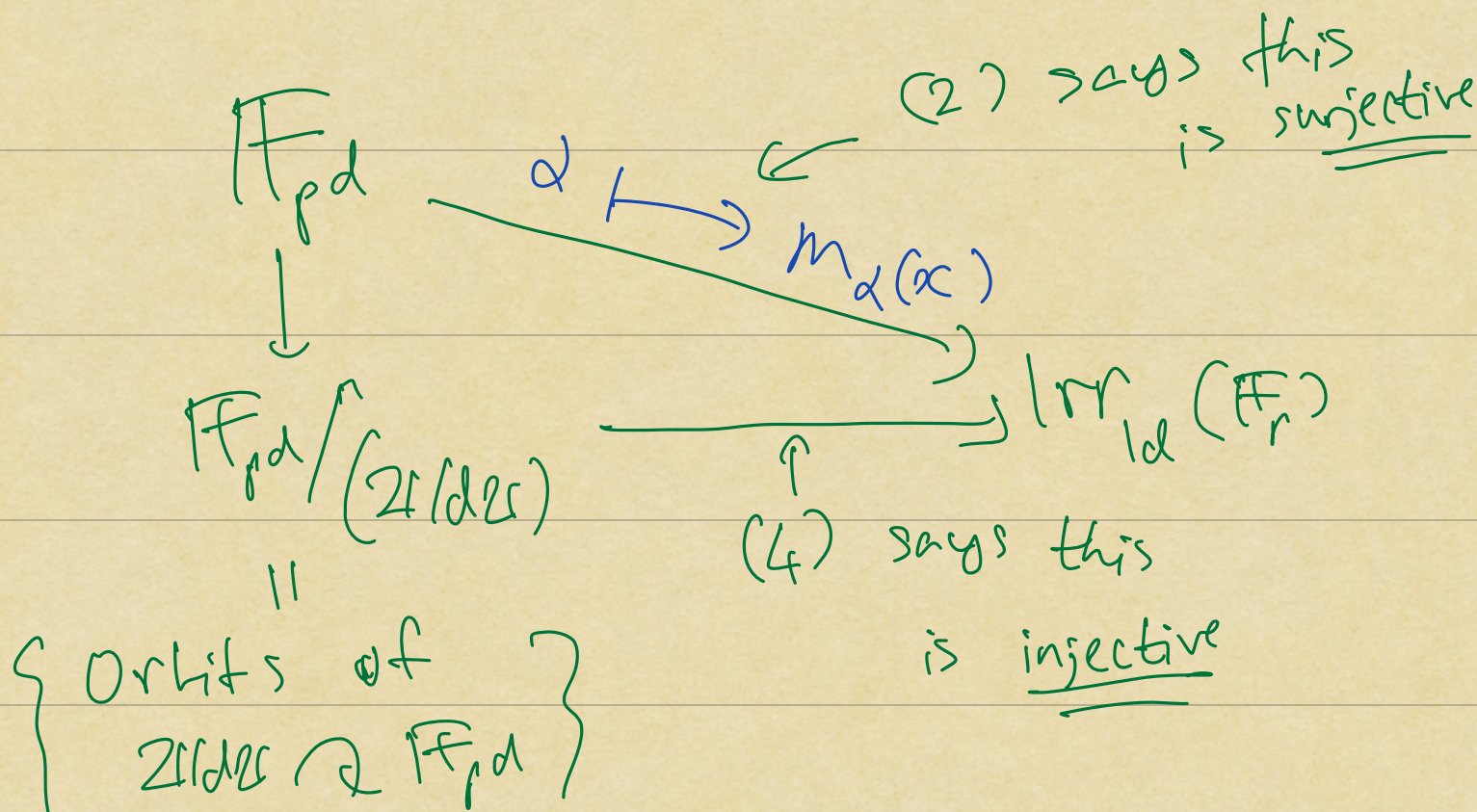
$$(\bar{c}, \alpha) \longmapsto \text{Fr}^{\bar{c}}(\alpha) = \alpha^{p^{\bar{c}}}$$

$$\text{irr}_{\text{Id}}(\mathbb{F}_r) \stackrel{\sim}{=} \left\{ \begin{array}{l} \text{Orbits of} \\ \mathbb{Z}/d\mathbb{Z} \curvearrowright \mathbb{F}_{p^d} \end{array} \right\}$$

$\cup$

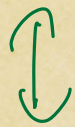
$$\begin{array}{c} f(x) \\ \deg r \mid d \end{array} \longmapsto \left\{ \begin{array}{l} \text{zeros of} \\ f(x) \text{ in } \mathbb{F}_{p^d} \end{array} \right\}$$

$$\left\{ \underset{\substack{\uparrow \\ \text{one} \\ \text{zero}}}{\alpha}, \text{Fr}(\alpha), \text{Fr}^2(\alpha), \dots \right\}$$



Notation:

$k \leq L$: field extension



L/k

NOT A QUOTIENT!

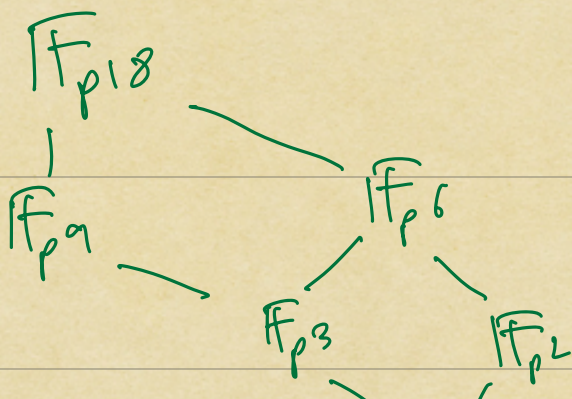
$$\text{Aut}(L/k) = \left\{ \varphi: L \xrightarrow{\sim} L \text{ field auto.} \right. \\ \left. \text{s.t. } \varphi(a) = a, \forall a \in k \right\}$$

Fact:

$$\mathbb{F}_{p^r} \leq \mathbb{F}_{p^d} \Leftrightarrow r \mid d$$

$$(\Leftarrow) k = \mathbb{F}_{p^r} : k_r = \{ \alpha \in \mathbb{F}_{p^d} : \underbrace{\text{Fr}^r(\alpha) = \alpha} \}$$

$$\text{then } |k_r| = p^r.$$



Linear transformations

Defⁿ k : field

V, W : k -vector spaces

A k -linear transformation

$$T: V \longrightarrow W$$

is a homomorphism of abelian groups s.t. $\forall a \in k, v \in V$

$$T(a \cdot v) = a \cdot T(v)$$

T is an isomorphism of k -vector spaces

if it is bijective

Altn: T is invertible

Rmk: If T is an isomorphism,

then $T^{-1}: W \rightarrow V$

is also an isomorphism of
 k -vector spaces

Pf: Need to check that

T^{-1} is a k -linear transformation

$$\bullet \forall w_1, w_2 \in W$$

$$T^{-1}(w_1 + w_2) \stackrel{?}{=} T^{-1}(w_1) + T^{-1}(w_2)$$

$$\downarrow T(\cdot)$$

$$w_1 + w_2 \stackrel{?}{=} T(T^{-1}(w_1) + T^{-1}(w_2))$$

$$T \text{ is a linear transformation} \rightarrow \boxed{=} T(T^{-1}(w_1) + T^{-1}(w_2)) \\ \stackrel{\checkmark}{=} w_1 + w_2$$

$$\bullet \forall \alpha \in k, w \in W$$

$$T^{-1}(a \cdot w) \stackrel{?}{=} a \cdot T^{-1}(w)$$

$$\hookrightarrow T(\cdot)$$

$$a \cdot w \stackrel{?}{=} T(a \cdot T^{-1}(w))$$

$$\boxed{a \cdot T(T^{-1}(w))} \\ \neq a \cdot w$$

Example:

$$\textcircled{1} \quad T: V \longrightarrow W \quad : \quad \text{zero or trivial} \\ v \longmapsto 0$$

$$\textcircled{2} \quad \text{id}_V: V \longrightarrow V \quad : \quad \text{identity} \\ v \longmapsto v$$

$$\textcircled{3} \quad D: k[x] \longrightarrow k[x] \\ f(x) \longmapsto f'(x)$$

$$\sum a_i x^i \longmapsto \sum i a_i x^{i-1}$$

$$(4) \quad T: k^n \longrightarrow k^m$$

Within k^n , we have

the standard basis vectors

$$e_1 = (1, 0, \dots, 0)$$

$$e_2 = (0, 1, \dots, 0)$$

!

$$e_n = (0, \dots, 0, 1)$$

Moreover, every vector $\overset{v}{\underset{\parallel}{(a_1, a_2, \dots, a_n)}}$
 $\in k^n$

is equal to a linear combination

$$v = a_1 e_1 + a_2 e_2 + \dots + a_n e_n$$

$$\downarrow T$$

$$\downarrow T$$

$$T(v) = a_1 T(e_1) + a_2 T(e_2) + \dots + a_n T(e_n)$$

Therefore, T is determined
entirely by the vectors

$$T(e_1), T(e_2), \dots, T(e_n)$$

Observation

(Homomorphism property of K^n)

Giving a K -linear transformation

$$T: K^n \longrightarrow W \leftarrow \text{any } K\text{-vector space}$$

is equivalent to giving

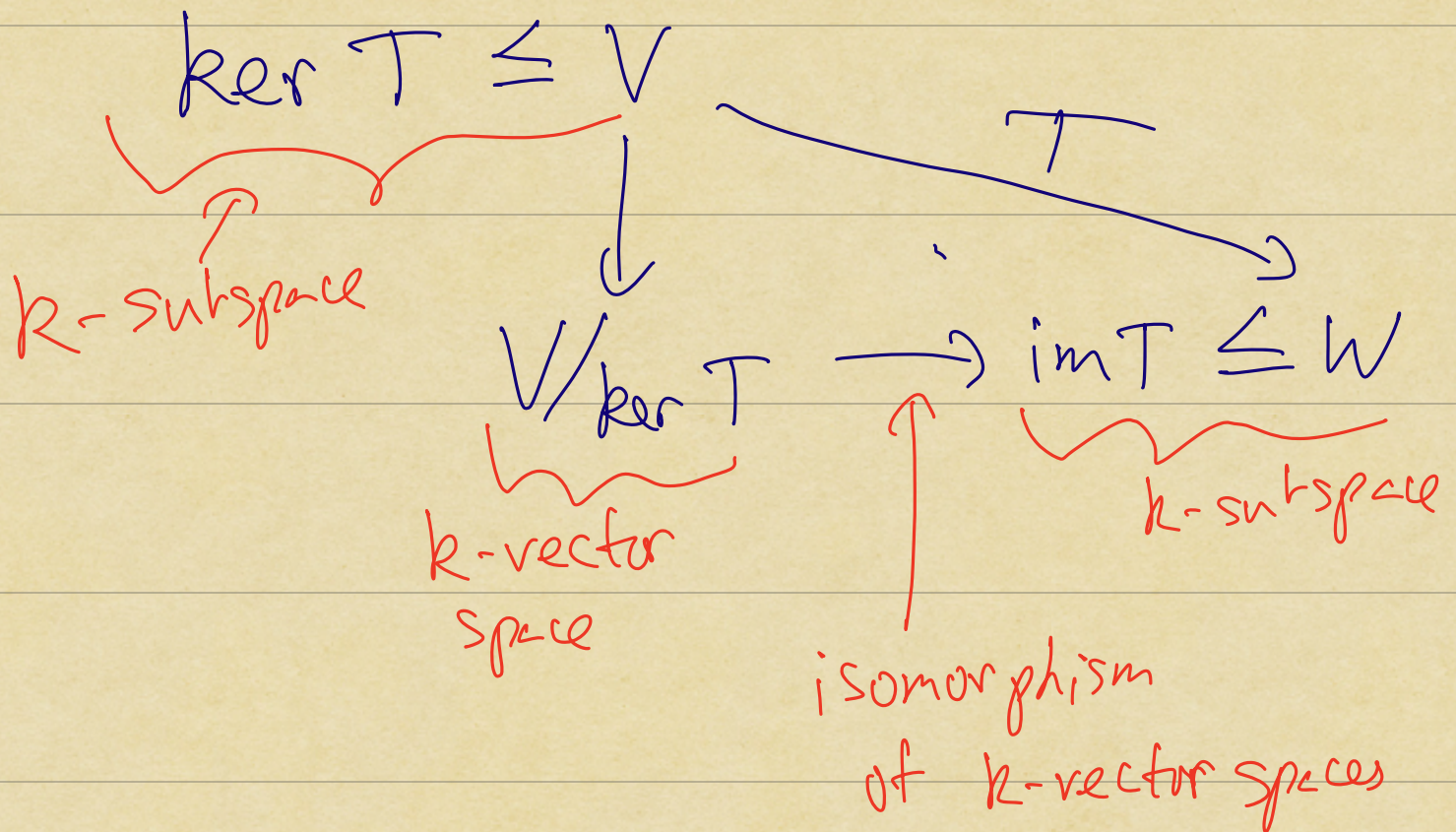
n vectors

$$\begin{array}{ccccccc} T(e_1) & , & T(e_2) & , & \dots & , & T(e_n) \in W \\ \parallel & & \parallel & & & & \parallel \\ w_1 & & w_2 & , & \dots & , & w_n \end{array}$$

T is given by

$$(a_1, a_2, \dots, a_n) \mapsto a_1 w_1 + a_2 w_2 + \dots + a_n w_n \in W$$

Defⁿ $T: V \rightarrow W$ k -linear transformation



Rmd(1)

$W \leq V$: k -subspace

π

V/W

is a k -linear transformation

Pf: Need to check

$$a \cdot (v + w) = a \cdot v + w$$

is well-defined

$$w \in W \Rightarrow a \cdot w \in W$$

$$a \cdot (v + w + W) = a \cdot v + a \cdot w + W$$

$\parallel$

$\parallel$? ✓

$$a \cdot (v + w) = a \cdot v + w$$

Observation

$$T: k^n \longrightarrow W$$

$$\downarrow$$
$$w_1, w_2, \dots, w_n$$

$$T(a_1, \dots, a_n) = \underbrace{a_1 w_1 + a_2 w_2 + \dots + a_n w_n}_{k\text{-linear combinations}}$$

Then

$$\text{im } T = \langle \{w_1, w_2, \dots, w_n\} \rangle_k$$
$$\leq W$$

Defⁿ W is a finitely generated k -vector space if the following equiv. conditions hold:

(1) $\exists$ finite set
 $X \subseteq W$
s.t. $\langle X \rangle_k = W$

(2) $\exists$ a surjective k -linear transformation

$$T: k^n \longrightarrow W$$

for some $n \geq 1$.